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Project Report

A Bilingual Recipe Assistant Combining a Knowledge Graph, Lightweight NLP, and LLMs

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I. ABSTRACT

We present a bilingual (PT/EN) recipe assistant that combines a curated recipe Knowledge Graph (KG), intent classification, entity extraction and fuzzy linking, SPARQL-based retrieval, and large language models (LLMs) for summarization. The system supports ingredient, tag, and time-constrained queries, and integrates a React/Vite frontend with a Python backend. We describe the data curation, approach, implementation, and experiments, highlighting strengths, limitations, and future directions.

II. INTRODUCTION

Motivation. Recipe search is often unstructured and multilingual. Users query for ingredients, dietary tags, or time constraints (e.g., “receitas em menos de 30 minutos”). Most systems either rely on pure keyword search or opaque large language model answers. We aim to combine structured knowledge graph reasoning with neural text generation, ensuring factual retrieval and multilingual usability.

Problem Description. Given a user query in Portuguese or English, the system must:

- detect intent;
- extract and link entities to KG resources;
- retrieve candidate recipes via SPARQL or semantic similarity;
- generate bilingual summaries;
- present results through an interactive interface.

Goal and Approach. Our solution includes:

- a curated recipe knowledge graph in RDF/Turtle;
- a FAISS embedding index for recipe descriptions;
- an intent classifier based on BERT;
- NLP pipelines for entity extraction and fuzzy linking;
- LLM-based bilingual summarization;
- a React/Vite-based frontend.

III. RELATED WORK

Structured recipe recommendation has been explored using ontologies and knowledge graphs. Approximate string

matching is commonly applied for entity linking. Retrieval-Augmented Generation (RAG) improves factuality by grounding generation in external data sources. Multilingual systems rely on language detection and selective translation. Our system integrates these ideas into a bilingual KG-driven assistant.

IV. DATA AND APPROACH

A. Data

The dataset includes:

- `recipes_graph_cleaned.ttl`: knowledge graph of recipes and ingredients;
- curated CSV files for analysis;
- vector embeddings stored via FAISS.

B. Pipeline

- 1) Intent classification (BERT);
- 2) Language detection and normalization;
- 3) Entity extraction and fuzzy linking;
- 4) Time parsing with regex;
- 5) KG/SPARQL retrieval and similarity search;
- 6) LLM-based summarization.

V. IMPLEMENTATION

A. Backend

Implemented in Python, including:

- Intent classification module;
- NLP pipeline with spaCy and RapidFuzz;
- SPARQL querying layer;
- FAISS-based similarity search;
- LLM interface (Groq and Ollama).

B. Frontend

The user interface is based on React/Vite:

- Markdown rendering;
- Language selection;
- Animated suggestions and message flow.

VI. EXPERIMENTATION

A. Setup

We evaluated PT/EN queries by ingredients, tags, and time constraints.

B. Results

Positive cases showed accurate intent routing and bilingual output. Initial failures in time parsing and API configuration were mitigated through relaxed matching and environment validation.

C. Metrics

- Intent accuracy (placeholder);
- Entity linking precision@k (placeholder);
- Time extraction success rate (placeholder);
- Human quality judgments for bilingual summaries (placeholder).

VII. CONCLUSIONS AND FUTURE WORK

We demonstrated that combining symbolic reasoning with neural generation produces accurate and fluent responses. The bilingual KG-powered assistant effectively supports multiple query types.

Future work includes:

- expanding KG coverage;
- adding automatic evaluation metrics;
- improving translation robustness;
- integrating structured outputs.

REFERENCES